

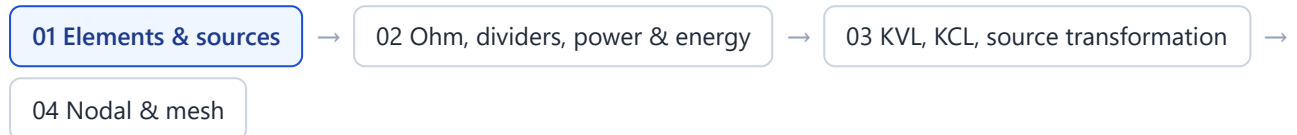
Circuits & Systems — Study Pack (MTE)

ECE2107 · B.Tech ECE Sem 3 · follows the professor's MTE syllabus (dated 10/9/26) · built from two sets of class notes · every answer checked by circuit simulation

The path through this pack

The order is the professor's MTE syllabus, item by item.

Part I — elements, laws, analysis



Part II — theorems (DC and AC)



Part III — transient analysis



Each file needs the ones before it. 04 is KCL/KVL (03) written as a method. 06 uses 05's "switch a source off" rule, 04's nodal analysis and 01's dependent sources. 07 is built on 06. 08 uses 06 for the time constant; 09 uses 08's Laplace route.

File	Syllabus item	What it teaches	Board Qs
01	1 · elements (R, L, C, IDS, DS)	active vs passive, independent and dependent sources (with devices), R, L, C laws, what can't jump, lumped vs distributed	—
02	2 · laws	Ohm, voltage and current division (DC and AC), power sign convention, energy, ladders	4
03	3 · KVL, KCL, source transformation	the two laws, power balance, source transformation	1 + 1 textbook
04	4 · nodal and mesh	both methods, with current sources and dependent sources	1 + 1 textbook
05	5 · superposition	method, the cases where it fails, AC sources at different frequencies	2 + 1 textbook
06	5 · Thévenin & Norton	with a dependent source; with AC impedances	1 + 1 textbook
07	5 · maximum power transfer	$R_L = R_{Th}$, efficiency, conjugate match (worked)	2 sets + 1 textbook
08	6 · first order	$0^- / 0^+ / \infty$, τ , the one formula, Laplace, test signals δ , u , r	4 + 2 textbook
09	7 · second order	series RLC: over-, critically, under-damped, undamped	textbook
10	all	one paper: 12 theory prompts + 20 numericals	14 + 6 textbook

Where the questions come from

A question that opens a section is **quoted from the professor's board** where one exists, and tagged with its notes page: **Class p.7** = first notes set (the 23-page scan), **Set 2 p.9** = second notes set (22 photos). Where the syllabus names a topic the class has not yet worked a question on, the pack uses a standard textbook-style problem, tagged **textbook**.

	Base	Status
P1	The professor's board questions (notes sets 1 and 2)	14 questions , all used
P2	Textbook problems	7 , written for syllabus items with no board question (KVL, nodal/mesh, AC theorems, Laplace, test signals, RLC). Values chosen so answers are clean; all checked by simulation

Marks per question: **not known** (no hand-out, no past paper yet).

How each file runs

Part	What happens
Map	the file's pieces, in order
Build	each step opens with a real question → you guess → the answer teaches the idea → a Check
Exam form	what to reproduce in the exam: statements, procedures, the formula sheet
Traps	the mistakes that cost marks
Self-test, Answers	answers are on the last page(s) of every file

THE ONE RULE THAT MAKES THIS WORK

When you reach a **Guess first** box, write a guess on paper **before** reading on: one line, however wrong. A wrong guess that gets corrected sticks better than a right answer you only read.

Mark every Check and Self-test item:

- **clean** → revisit in 3 days
- **with struggle** → revisit tomorrow
- **wrong** → redo today, after the file

Units and symbols used everywhere

- Uppercase **V, I** = DC (constant) values. Lowercase **v, i** = values that change with time. Complex values such as $5 - j5$ V are **phasors**: amplitude and phase in one number.
- Resistances in **kΩ** with currents in **mA** give voltages in **V** directly: $2 \text{ k}\Omega \times 3 \text{ mA} = 6 \text{ V}$. Most questions use this.
- **R_{Th}, V_{Th}** = Thévenin resistance and voltage. **I_N** = Norton current. **R_L** = load resistance.
- **0⁻** = just before the switch moves. **0⁺** = just after. **∞** = long after (steady state).

Where the two notes sets disagree with the pack

Topic	Notes say	Pack says	Why
Superposition, two 5 V sources (05 §5)	set 2: $0 + 0 = 0$ A, "LIMITATION"; set 1: 10 mA	0 A by superposition, true 5 mA	set 2 is the correct use of the rule and shows why it fails; the pack follows set 2
Thévenin with 2I (06 §2)	set 2: $V_{Th} = 5.0025$ V	20/3 V	set 2 reads 2I as $2 \Omega \times I$; the pack reads $2 \text{ k}\Omega \times I$. Both are right for their reading. Ask the professor
RL switch that opens (08 §5)	$I_R(0^+) = 0$, $I_L(0^+) = 10$ mA	cannot both hold in one series loop	ask the professor which path the current takes

How this pack was checked

- Facts: checked against the textbook-sourced circuits knowledge base (Hayt, Van Valkenburg, Sudhakar & Shyammohan) and Alexander & Sadiku.
- Numbers: every circuit in this pack was solved a second time by computer (nodal analysis and time simulation), and the answers here match. The script is `src/verify_mte.py`.
- Where the class notes had a slip of the pen, the pack uses the corrected version and says so in a Note.